

Sri Lankan Black Pepper

Harvesting & Post-Harvest Processing

Knowledge Domain: 03 — Cultivation (Harvesting) and Processing

This document covers the harvesting practices and post-harvest processing workflow for black pepper (*Piper nigrum*) in Sri Lanka. It provides knowledge on when and how to harvest, and how raw green berries are transformed into the market-ready dried black pepper product.

1. Harvesting Indicators

Harvesting at the correct maturity stage is the single most important factor in determining the quality, colour, and pungency of the final black pepper product. Harvesting too early or too late both result in significant quality losses.

1.1 Maturity and Visual Cues

- **Time to Maturity:** Black pepper typically reaches full harvest maturity 7 to 8 months after flowering. Farmers should mark flowering dates to anticipate the harvest window.
- **Primary Visual Signal:** The correct moment to harvest is when one or two berries on a spike begin to turn bright orange or red. This colour change signals that the entire spike has reached the optimal piperine content.
- **Common Mistakes:** Do not wait for the whole spike to turn red — this causes overripe berries to drop and reduces yield. Early-stage green spikes harvested too soon will produce wrinkled, low-quality black pepper.
- **Harvesting Method:** Hand-picking is the standard harvesting method in Sri Lanka. Farmers use ladders or tall stools to access spikes on support trees (*Gliricidia*) that can reach 4 metres.

Harvest Readiness Reference:

Indicator	Target Value	Practical Notes
Time from Flowering	7 to 8 months	Monitor from first spike emergence
Berry Color (Harvest Signal)	1–2 berries turn bright orange or red	Do not wait for full spike ripening
Berry Texture	Firm, plump, fully developed	Shriveled or pale berries indicate premature harvest
Spike Condition	Majority of berries still green	Green berries produce the best black pepper quality

Tip: Stagger harvesting across the farm — not all vines ripen simultaneously. Plan 2–3 rounds of harvesting over the season to capture peak-quality spikes from all vines.

2. Post-Harvest Processing

Post-harvest processing converts the raw freshly harvested green berries into stable, dried black pepper. Each step must be performed promptly and correctly to preserve quality, meet export standards, and prevent spoilage.

2.1 Full Processing Workflow

The complete post-harvest processing chain from spike to market is summarised below:

Processing Step	Input	Output	Key Action / Notes
1. Threshing	Freshly harvested spikes	Loose green berries	Separate berries from rachis (stalk) manually or by thresher
2. Blanching	Loose green berries	Uniform dark green/black skin	Immerse in hot water (80°C) for 1–2 minutes. Optional but recommended for export quality.
3. Sun Drying	Blanched or raw green berries	Dry black pepper (≤12% MC)	Spread on bamboo mats or black polyethylene sheets; 3–5 days under direct sunlight
4. Cleaning/Sorting	Dried black pepper	Graded clean pepper	Remove stalks, debris, shrivelled, and discoloured berries
5. Storage	Graded pepper	Market-ready product	Store in clean, dry, well-ventilated jute or polypropylene bags; moisture < 12%

2.2 Threshing (Spike Removal)

Immediately after harvesting, the green berries must be separated from the spike stalks (rachis). Delayed threshing accelerates fermentation and spoilage.

- **Setup:** Spread harvested spikes on a clean, dry cement floor or tarpaulin.
- **Method:** Thresh manually by rubbing spikes between palms or feet, or use a mechanical thresher for larger volumes.
- **Cleanup:** Collect and remove all rachis material after separation to avoid contamination of the final product.

2.3 Blanching — The Key to Premium Ceylon Black Pepper

Blanching is an optional but highly recommended step for farmers targeting premium-grade or export-quality "Ceylon Black Pepper." It is the key differentiator in producing a uniformly black, visually appealing product.

- **Preparation:** Fill a large vessel with clean water and heat to exactly 80°C (avoid boiling, which damages the berries).
- **Blanching Process:** Immerse the separated green berries in the hot water for 1 to 2 minutes only. Over-blanching softens berries and damages quality.
- **After Blanching:** Remove berries immediately and spread them on drying surfaces — blanched berries must not be left in standing water.

Benefits of Blanching:

- 1. Produces a uniform, deep black colour across the entire batch — essential for export premium pricing.
- 2. Accelerates the subsequent drying process by breaking down the outer skin, allowing faster moisture escape.
- 3. Kills surface microorganisms and pathogens, making the final product safer and extending shelf life.

Export Note: *Blanched and properly dried Ceylon Black Pepper commands higher prices in international markets due to its consistent colour, lower microbial load, and superior appearance.*

2.4 Drying

Drying is the most critical post-harvest step for long-term quality preservation. Insufficient drying causes mould growth, mycotoxin contamination, and complete product loss.

- **Sun Drying Setup:** Spread berries in a single layer on bamboo mats, clean cement floors, or black polyethylene sheets under direct sunlight.
- **Turning:** Turn the berries every 2–3 hours to ensure uniform drying on all sides.
- **Duration:** In Sri Lanka's tropical climate, sun drying typically takes 3 to 5 days to achieve the target moisture content.
- **Target Moisture:** The final moisture content must be 12% or below to prevent mould growth and ensure quality is maintained during storage and transport.
- **Alternative:** In wet or cloudy weather, mechanical dryers operating at 50–60°C can be used as an alternative to sun drying.

Drying Standards Reference:

Parameter	Standard / Guideline
Drying Surface	Bamboo mats, clean cement floors, or black polyethylene sheets
Drying Duration (Sun)	3 to 5 days under direct tropical sunlight
Target Moisture Content	12% or less (mandatory for safe storage and export)
Artificial Drying (Alternative)	Mechanical dryers at 50–60°C if weather is unfavourable
Turning Frequency	Every 2–3 hours to ensure even drying

Warning: *Never pile berries in thick heaps during drying — this traps heat and moisture, promoting rapid mould development. A single-layer spread is mandatory for safe drying.*

3. Quality Standards and Storage

Once dried, black pepper must be cleaned, graded, and stored correctly to preserve quality until it reaches the buyer or export facility.

3.1 Cleaning and Sorting

- **Debris Removal:** Manually or mechanically remove all stalks, dust, stones, and plant debris from the dried batch.
- **Berry Grading:** Discard shrivelled, discoloured, or undersized berries as these reduce overall grade and market value.
- **Sizing:** Grade the final product by size using mesh screens or sieves to separate bold, medium, and small berries for different market channels.

3.2 Storage Conditions

- **Bags:** Store in clean, dry, well-ventilated jute or polypropylene bags. Avoid airtight plastic containers, which trap humidity.
- **Environment:** Maintain storage room temperature below 30°C with relative humidity below 70%.
- **Stacking:** Keep bags elevated off the floor on pallets to prevent moisture absorption from the ground.
- **Monitoring:** Inspect stored pepper monthly for signs of mould, insect infestation, or unusual odour.

DEA Guideline: *The Department of Export Agriculture recommends testing moisture content with a calibrated moisture meter before bagging. Batches exceeding 12% moisture must be re-dried before storage.*